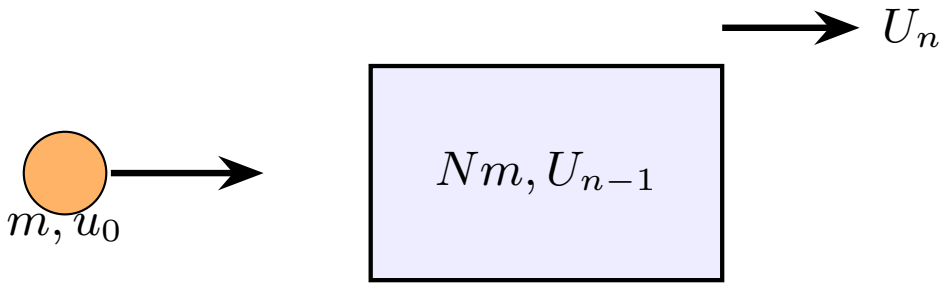
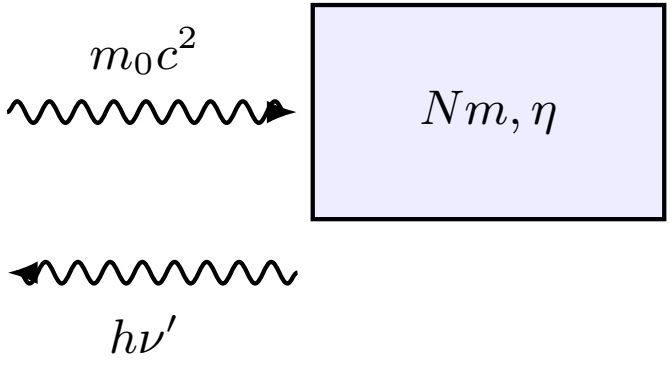


$$U_n = \frac{N-1}{N+1}U_{n-1} + \frac{2}{N+1}u_0$$



$$e^{\eta'} = e^{\eta} + \frac{2m_0}{Nm}$$